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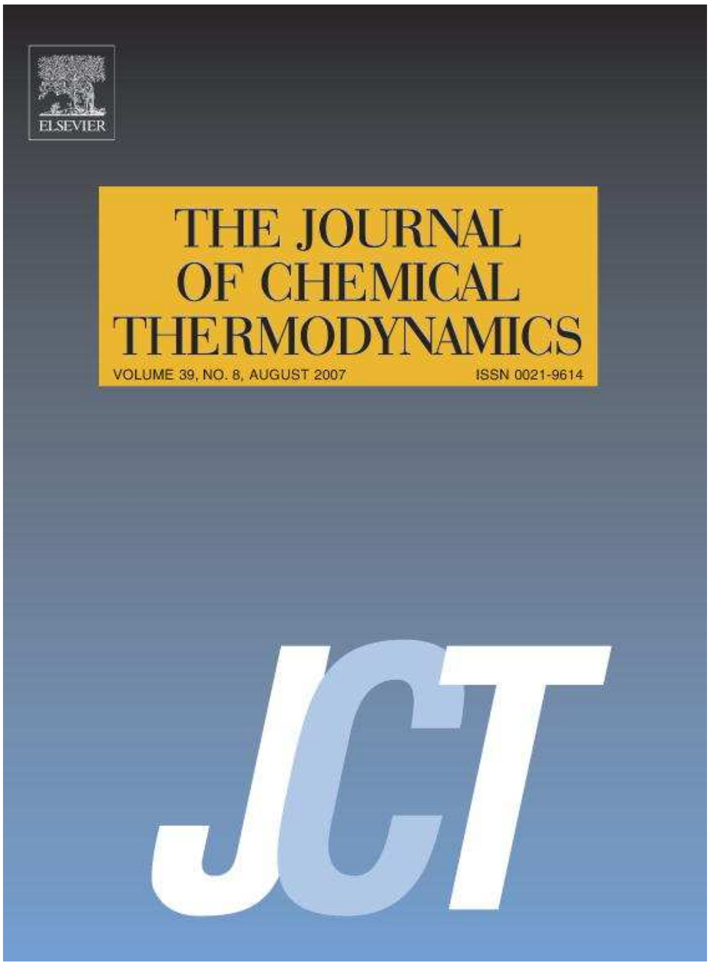
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Accurate density measurements for a 91% methane natural gas-like mixture

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Abstract

Accurate density data are necessary to validate equations of state (EOS) used in custody transfer of natural gas. The American Gas Association (AGA) AGA8-DC92 EOS, currently the industry standard, has been validated against a databank of gas mixtures composed mostly of binary data. Accurate density data for multicomponent mixtures are necessary to check the ability of AGA8-DC92 to cover the entire range of pressure, temperature, and compositions encountered in custody transfer. This paper uses a magnetic suspension densimeter to measure the densities of mixtures containing nine components with *n*-pentane being the heaviest. We have used a high pressure, high temperature, compact single-sinker magnetic suspension densimeter to measure densities of a simulated natural gas mixture having 91% methane composition after validating its operation by measuring densities of pure argon, nitrogen, and methane in the range (270 to 340) K and (3.447 to 34.474) MPa. The agreement with the reference EOS for the pure components is about 0.03%.

The estimated 2σ uncertainty of the densities after compensating for apparatus and fluid effects is 0.12% with 93% of that value coming from the composition uncertainty. The measured data indicate the advisability of performing an extensive study of the AGA8-DC92 EOS for natural gas mixtures.

Keywords: Densities; Phase loops; Natural gas; Magnetic suspension densimeter; Isochoric apparatus

1. Introduction

Given increased activity in petroleum, cryogenic, and natural gas businesses, trustworthy estimates of thermodynamic properties and accurate experimental density measurements are necessary to design engineering processes. Accurate prediction of thermophysical properties is an essential requirement in optimum design and operation of most process equipment involved in petrochemical production, transportation, and processing. More importantly, the oil and natural gas industry requires high accuracy phase equilibrium and pressure–density–temperature (P, ρ, T) data at several stages between exploration and market end use. Knowing the temperature, pressure, and composition, an accurate density can be calculated by using a proper EOS. However, the accuracy of such an EOS in turn depends upon the accuracy of the experimental data used to establish it. For this reason, to develop a reference quality EOS, reliable thermodynamic property measurements of the fluids must be available. The motivation for measuring a 91% methane gas mixture is the need of confirmation for the existing industry benchmark EOS developed by Starling and Savidge [1], AGA8-DC92, which has been validated against a databank of experimentally measured compressibility factors for pure components, binary mixtures, and some multi-component gas mixtures. At Texas A&M University, we have assembled a state-of-the-art, high pressure, high temperature, single-sinker magnetic suspension densimeter (MSD) for accurate density measurements.

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Vibrating tubes and vibrating forks are common density measuring techniques. These devices measure the fluid density of interest by determining the oscillation frequency of the vibrating element in the fluid. This instrument provides accurate results in very little time. However, frequent calibration is necessary for this apparatus to maintain its accuracy. Moreover, when the density of the fluid is vastly different from air or pure water (frequently used as reference fluids because of well known thermophysical properties) the uncertainty of the measurements increases as reported by Kuramoto *et al.* [2].

Another well-established and widely used density measurement apparatus is the Burnett apparatus. Burnett [3] suggested a technique to measure the densities of sample fluids without measuring the mass or volume directly. Only pressure and temperature are measured before and after expansion of the sample from a single volume into the combination of the original volume and a second volume. Wagner *et al.* [4] note the well known but serious problem associated with adsorption of the sample gas on the inner surfaces of the Burnett cells. Also a Burnett apparatus is difficult to automate fully because of frequent valve operation.

2. Apparatus

The hydrostatic buoyancy force technique based upon the classic Archimedes' principle states that immersing a solid body (sinker) in a fluid displaces a volume of fluid the weight of which equals the buoyancy force exerted by the fluid upon the sinker. The buoyancy force in turn is proportional to the density of the fluid in the measuring cell under pressure. However, Kleinrahm and Wagner [5] note that zero point shifts of the balance, buoyancy forces on auxiliary devices, adsorption effects, and surface tension may reduce the accuracy of such measurements.

To overcome limitations in achievable accuracy, the need for frequent calibration of the apparatus with reference fluids, complexity of operation, limitations on temperature and pressure, and force transmission error, Kleinrahm and Wagner [5] have introduced an MSD based upon magnetic levitation of the sinker in the measuring cell. The novelty of the magnetic suspension coupling is that it uses non-physical-contact force transmission between the sinker in the pressurized cell and the weighing balance at atmospheric pressure, thus allowing a cell design that covers a wide temperature and pressure range, as noted by Lösch [6].

Kleinrahm and Wagner [5] modified the hydrostatic buoyancy force method by introducing an alternative force transmission method, in which they levitated two sinkers through a magnetic suspension coupling. By compensation for surface tension, buoyancy, adsorption effects, and shifts in zero-point of the balance, a two-sinker MSD improved the accuracy of the density measurements.

Operation of a two-sinker MSD is rather complex and its advantages are not required for the medium or high-density measurements encountered in many practical applications. To extend the instrument range towards higher temperatures and pressures, Wagner *et al.* [4] have developed a single-sinker densimeter. Although the single sinker design is simpler than that of the two-sinker densimeter, it is still possible to perform accurate density measurements at relatively low gas densities by applying some of the features of the two-sinker principle, as noted by Lösch *et al.* [7]. The single-sinker densimeter also operates based upon Archimedes' principle, and the force transmission comes from levitation of the sinker the high-pressure cell. The density results from measuring m_v, the 'true mass' of the sinker in vacuum, m_a, the 'apparent mass' of the sinker in the presence of fluid and V_s, the volume of the sinker:

$$\rho = (m_v - m_a)/V_s \tag{1}$$

Wagner *et al.* [4] have concluded that the accuracy of density measurement from a single-sinker densimeter is lower than that from a two-sinker densimeter especially at low densities because it lacks compensation for adsorption effects. Also, the force transmission error has more effect upon total density uncertainty than with a two-sinker densimeter. For lower densities, having the load compensation system outside the measuring cell, as with the single-sinker design, is less effective than having it inside the cell, as with the two-sinker design. They also conclude that the accuracy of density measurements from a single-sinker densimeter is better than that from a Burnett apparatus because the adsorption is a point error not a cumulative error.

In our laboratory, we use a compact single-sinker MSD manufactured by Rubotherm Präzisionsmesstechnik GmbH. The apparatus has an accuracy specification from the manufacturer of $\pm$(0.03% or 0.005 kg/m^3) for densities in the range (0 to 2000) kg/m^3 over a temperature range of (193 to 523) K and pressure range up to 200 MPa with a maximum pressure of 130 MPa at 523 K [9]. However, we conclude that, after including the temperature, pressure, composition, apparatus, and fluid effect uncertainties, the 2σ uncertainty of the density measurements for this mixture is 0.12%. The primary source of error is the composition accounting for 93% of the total. DCG Partnership Ltd. synthesized the gas mixture gravimetrically and verified the composition using GC analysis. Obviously, the more components in the mixture, the more uncertain the composition becomes.

In this work, the high-pressure cell resided in a constant temperature, liquid bath that kept the cell isothermal during a run. The bath fluid was a 50:50 mixture by volume of ethylene glycol and distilled water that covered a temperature range of (248 to 373) K. A four-lead capsule platinum resistance thermometer (PRT) with a temperature range of (84 to 533) K manufactured by Minco Products, Inc. provided temperature measurements. We used oil-free, resonating quartz crystal, absolute pressure transducers (40 and 14) MPa from Paroscientific Inc. with NIST traceable calibrations for pressure measurements. The sinker in this experiment was a titanium cylinder with a volume of $6.74083\ \mathrm{cm}^3$ with $0.0034\ \mathrm{cm}^3$ uncertainty and a mass of 30.39157 g, both measured at 293.15 K and 0.1 MPa by the manufacturer.

3. Experimental

Figure 1 illustrates the principal of operation [10]. The main component of the single-sinker densimeter is the magnetic suspension coupling. This coupling allows transmission of force from the pressurized measuring cell to the balance at ambient conditions without physical contact. The coupling consists of an electromagnet with a soft iron core, a permanent magnet made from samarium cobalt ($SmCo_5$), a ferromagnetic sensor core and a control system. The electromagnet attaches to the balance via a hook beneath the balance and the permanent magnet connects to the sinker in the measuring cell with the coupling–decoupling device. The position sensor core attaches to the permanent magnet shaft via a thread. A thin wall separates the two magnets, and a PID controller generates gentle vertical movements of the permanent magnet thus coupling and decoupling the sinker with the permanent magnet.

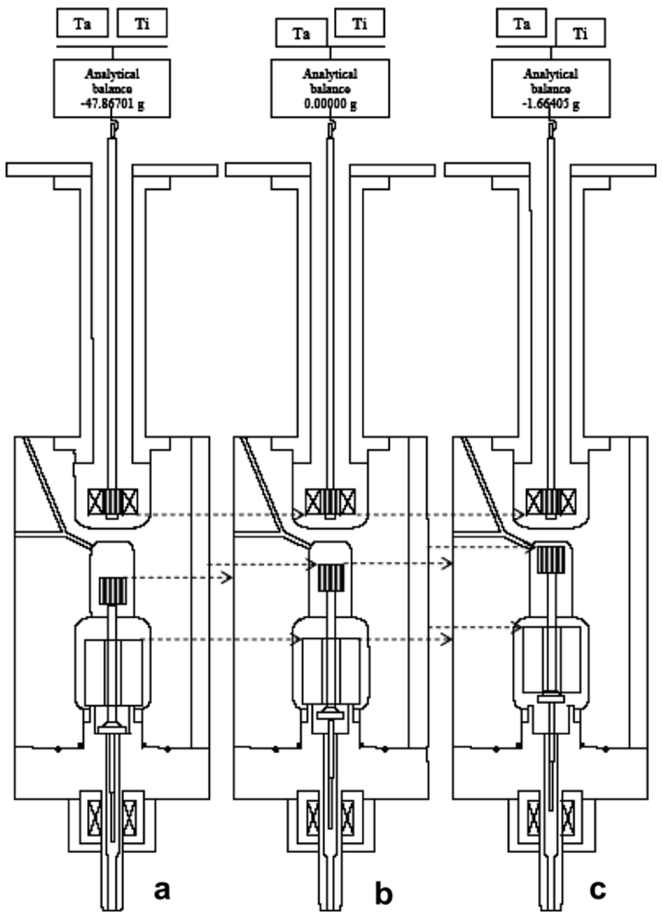

FIGURE 1. Operation of the MSA in vacuum using weight changing device (a) suspension control (SC) off, Ti and Ta both raised (b) SC on; zero point (ZP) position, Ta lowered, Ti raised (c) SC on; measurement point (MP) position, Ta raised, Ti lowered.

In the tare or zero position (ZP), the permanent magnet rises only a few mm such that it does not couple with the sinker, and the balance senses the weight of the electromagnet and its shaft, the permanent magnet and its shaft and the sensor. In the measurement position (MP), the electromagnet rises farther such that it couples with the sinker. The distance between the permanent magnet and the top of the pressure cell is (1 to 3) mm (depending upon gas density) in the measuring position and about 6 mm in taring position [2,11].

The balance has a load compensation system that allows operation near its zero point to improve the accuracy of the measurement. The technique employs cylindrical tantalum and titanium weights of nearly equal volume and surface area but with different masses, as suggested by Wagner *et al.* [4]. With the suspension in the zero position after taring the balance, we place the tantalum weight ($m_{\mathrm{Ta}} \approx 41.618$ g at ambient, $V_{\mathrm{Ta}} \approx 2.49\ \mathrm{cm}^3$ and $\rho_{\mathrm{Ta}} \approx 16{,}700\ \mathrm{kg/m}^3$) on the balance. Then, using a weight-changing device, we switch the tantalum weight with the titanium weight ($m_{\mathrm{Ti}} \approx 11.233$ g at ambient, $V_{\mathrm{Ti}} \approx 2.49\ \mathrm{cm}^3$ and $\rho_{\mathrm{Ti}} \approx 4500\ \mathrm{kg/m}^3$). In this measuring position, the sinker also has coupled with the balance, so the total weight is about 41.61 g. Because the balance readings are almost identical, the resultant reading at the measurement position is almost zero ($\sim$0.07 g). Based upon the measurement using the two external weights, we can re-write equation (1):

$$\rho = [(m_{\mathrm{V}} + m_{\mathrm{Ti}} - m_{\mathrm{Ta}}) - (m_{\mathrm{a}} + m_{\mathrm{Ti}} - m_{\mathrm{Ta}})]/V_{\mathrm{S}}(T,P) \qquad (2)$$

Now, as we increase the gas density, the balance reading moves farther from zero, but even at high density this reading is small compared to the total range of the balance. So, the application of the two external weights significantly reduces the errors of the balance caused by changes in the slope of the characteristic line [11]. Error propagation resulting from additional mathematical steps is not significant. Without the external weights, these errors would not cancel between ZP and MP.

In equation (2), the masses of Ta and Ti cancel between readings obtained at vacuum and at some pressure and temperature, so we assume that ambient conditions do not change during that time interval. Also, equal volumes of the Ta and Ti ensure that buoyancy effects of the ambient air are the same, and the equal surface area allows adsorption effects to cancel. The differential application of Archimedes' principal eliminates most of the major effects that reduce the accuracy of measurement. In addition, adsorption on the inner wall of the surfaces caused by surface roughness becomes insignificant [8].

Three effects, however, do not cancel. One is the interaction between the force transmitted and the Cu–Be cell, which is slightly diamagnetic. This means that the sinker mass is not the same if measured directly at the balance or measured through the coupling under the same conditions. This effect introduces a systematic, force-transmission error (FTE) on the order of 10^{-6} in the density [2]. Wagner *et al.* [11] have suggested a multiplying factor of $(1 \pm 2 \cdot 10^{-5})$ to the numerator in equation (1). However, using calibrated weights, Kuramoto *et al.* [2] have suggested a correction factor C_{housing} at vacuum defined as the ratio of the sinker mass measured with the coupling to that measured directly with the balance.

$$C_{\text{housing}} = m_{\text{direct}}/m_{\text{coupling}} \tag{3}$$

For our case the value of C_{housing} is 0.9999462. The second effect is a FTE caused by magnetic interaction with the samples that might be paramagnetic or diamagnetic. This effect persists for the same reasons cited in the previous paragraph. Klimeck *et al.* [8] formulate an empirical equation to correct this error that is valid for some diamagnetic pure fluids (methane, ethane, carbon dioxide, and nitrogen). Kuramoto *et al.* [2] correct the error in terms of a fluid dependent coefficient C_{liquid} using water and *n*-tridecane as reference fluids. However, they assume a constant difference of 5 mm between ZP and MP for all fluids, and McLinden *et al.* [12] show that the fluid effect on the FTE is negligible for such fluids. Finally, gas adsorption on the sinker surface affects the accuracy of measurement. However, this effect is significant only at low densities [4].

The uncertainty in the volume of the sinker is a major contributor to the uncertainty in the density measurement. By using an optical interferometer, Fujii [13] reduced the relative uncertainty for the volume of a steel sphere to as low as 10^{-11}. He used a silicon sphere instead of a cylinder or a cube because it is less susceptible to damage and spheres are available with excellent sphericities. The density calculated from the volume of a silicon sphere becomes a reference value to measure the density of solid samples (not necessarily spherical) using hydrostatic weighing methods.

The beryllium-copper cell (pressure tested to 300 MPa) contains a groove that allows insertion of a PRT. The top part of the cell has an axially bored cylindrical depres-

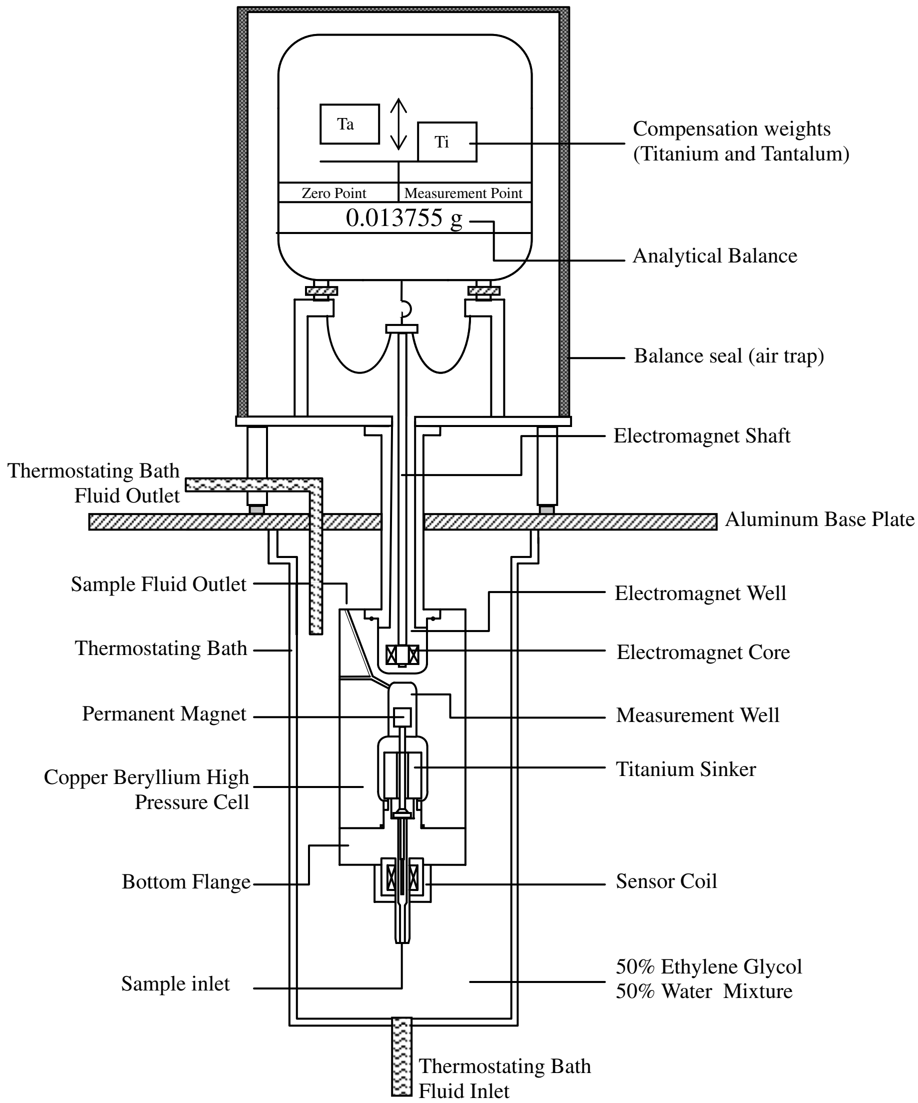

FIGURE 2. Magnetic suspension assembly.

sion containing the electromagnet suspended from the bottom hook of the weighing balance pan with a suspension tube. Figure 2 is a detailed sketch of the magnetic suspension assembly.

4. Results

Table 1 contains measured densities of the mixture over a temperature range of (270 to 340) K and a pressure range of (3.4 to 34.5) MPa and figure 3 presents the data as a plot. The composition of the mixture and the molar masses of the pure components appear in table 2. Zhou *et al.* [14] have measured the same mixture using an isochoric apparatus. However, their data are along isochors, and they do not correspond to those presented in this paper. The two sets of densities are consistent with each other within 0.05%

We have compared the measured molar densities to those predicted by the Hall–Yarborough (HY) [15], AGA8-DC92 and REFPROP [16] EOS. Lemmon *et al.* at NIST have developed REFPROP based upon the most accurate pure fluid and mixture models currently available. REFPROP implements three models for the thermodynamic properties of pure fluids: equations of state explicit in Helmholtz energy, the modified Benedict–Webb–Rubin equation of state, and an extended corresponding states (ECS) model. Mixture calculations employ a model that applies mixing rules to the Helmholtz energy of the mixture components and then a function to account for the departure from ideal mixing. We have used REFPROP version 7 in this paper. Figure 4 presents the relative deviations of the measured densities from the EOS. The measured data deviate by 1.25% from HY predictions, by 0.7% from the AGA8-DC92 predictions and by 0.25% from the REFPROP predictions.

Table 3 denotes the accuracy ranges for AGA8-DC92 predictions. Predictions in region 1 should be within 0.1% of data, region 2 within 0.3%, and region 3 within 0.5%. The predicted values clearly do not have such accuracies

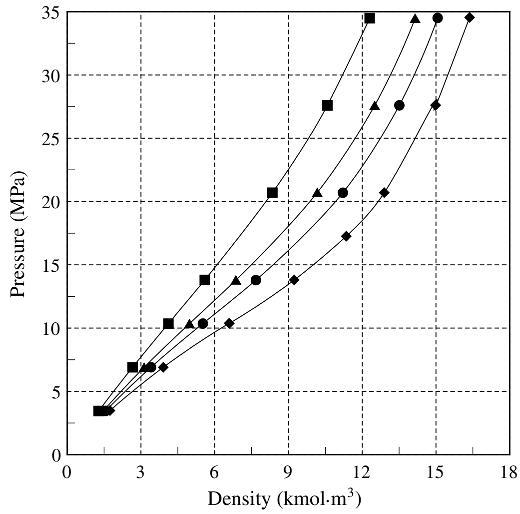

FIGURE 3. Measured density data for the mixture for different isotherms ■ experimental 340 K; ▲ experimental 305 K; ● experimental 290 K; ◆ experimental 270 K.

TABLE 1
Measured density values for the sample

Temperature (K)	Pressure (MPa)	Density/ (kg · m^3)
	270 K isotherm	
270.00	3.483	31.673
269.99	6.897	71.135
269.99	10.370	119.993
269.98	13.799	168.172
269.99	17.261	206.727
269.98	20.698	234.853
269.99	27.618	272.887
269.99	34.543	298.051
	290 K isotherm	
290.01	3.453	28.415
290.01	6.903	61.993
290.01	10.354	100.346
290.00	13.789	139.581
290.01	17.271	175.201
290.01	20.693	203.883
290.01	27.600	245.666
290.01	34.506	274.016
	305 K isotherm	
305.00	3.484	26.894
305.00	6.899	56.941
304.99	10.343	90.415
305.00	13.792	124.842
305.01	20.707	185.026
305.00	27.586	227.534
305.02	34.472	257.481
	340 K isotherm	
340.00	3.450	23.264
340.00	6.902	48.440
340.03	10.350	74.948
340.03	13.799	101.844
340.02	20.699	151.983
340.03	27.588	192.645
340.00	34.486	224.033

TABLE 2
Mixture composition and pure component properties

Component [*i*]	Actual composition [x_i]	MW/ (g · mol)	T_c (K)	P_c (MPa)
CH_4	0.90991	16.043	190.6	4.599
C_2H_6	0.02949	30.070	305.3	4.872
C_3H_8	0.01513	44.097	369.8	4.248
i-C_4H_{10}	0.00755	58.123	408.1	3.648
n-C_4H_{10}	0.00755	58.123	425.1	3.796
i-C_5H_{12}	0.00299	72.150	460.4	3.381
n-C_5H_{12}	0.00304	72.150	469.7	3.370
N_2	0.02031	28.014	126.2	3.400
CO_2	0.00403	44.010	304.2	7.383

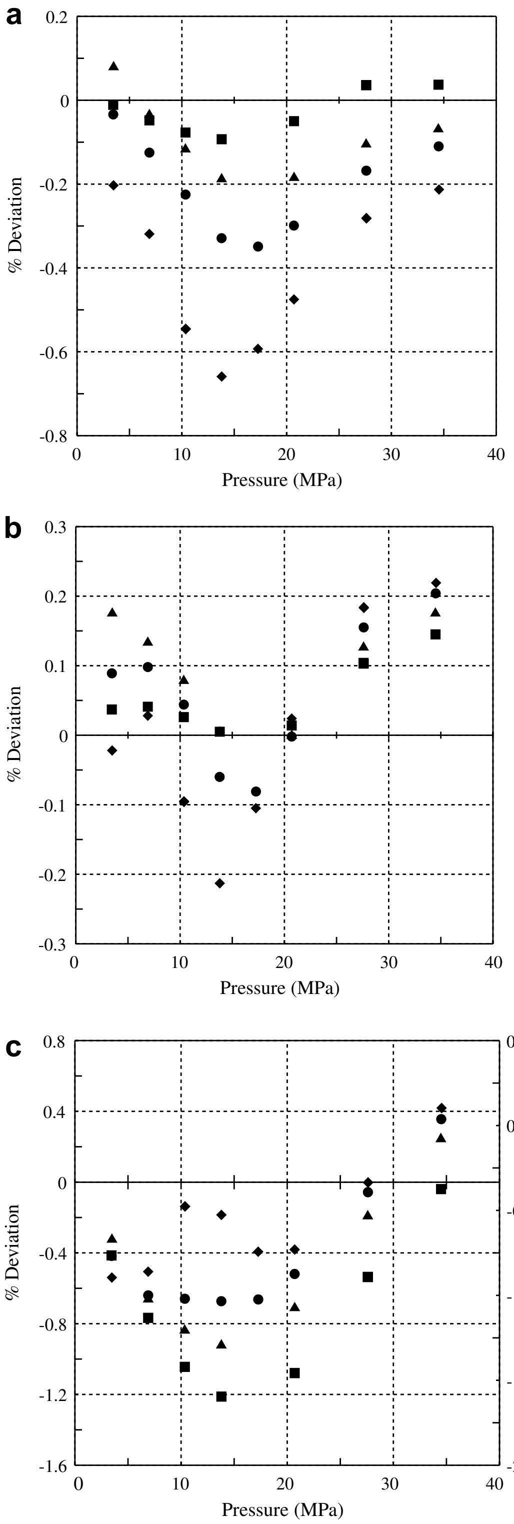

FIGURE 4. (a) Relative deviations between measured and AGA8-DC92 densities. (b) Relative deviations between measured and REFPROP densities. (c) Relative deviations between measured and HY densities ■ 340 K; ▲ 305 K; ● 290 K; ◆ 270 K.

for most of the data presented in figure 4. Figure 5 presents density deviations with respect to AGA8-DC92 and shows the uncertainty regions of AGA8-DC92.

TABLE 3
American Gas Association (1992) data regions

Data region	Minimum T (K)	Maximum T (K)	Minimum P (MPa)	Maximum P (MPa)
1	265	335	0	11.997
2	211	394	12.00	17.00
3	144	477	17.00	35.00

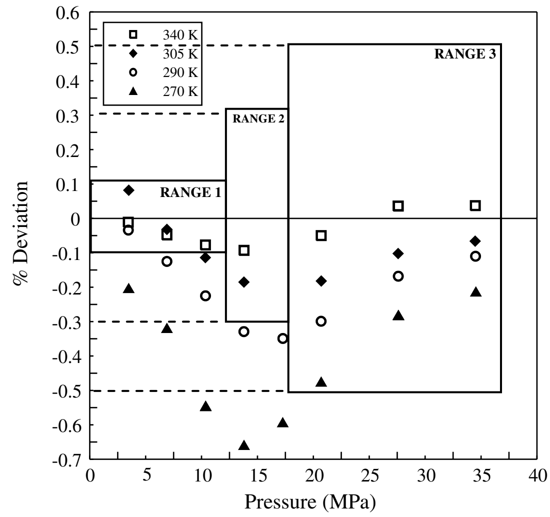

FIGURE 5. 91% sample density deviations w.r.t. AGA8-DC92 with AGA8-DC92 uncertainty regions.

In summary, the measured densities for the mixture deviate from the AGA8-DC92 predictions in regions 1 and 2 to a greater extent than expected but match in region 3. REFPROP predictions are generally better.

5. Conclusion

We have used a single-sinker magnetic suspension densimeter to measure the densities of a natural gas-like mixture with a composition in the normal range of AGA8-DC92. The AGA8-DC92 predictions have larger than expected deviations from our data. Predictions from REFPROP appear to be somewhat better while those from Hall–Yarborough are worse. The measured data indicate that AGA8-DC92, the natural gas custody transfer standard, may have problems in important regions of the phase space. AGA8-DC92 may require an extensive revision with more data covering multicomponent mixtures, or it may be necessary to develop a new equation of state for custody transfer purposes.

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